Empowering Real-World Sensing: Human-Centered Machine Learning for Imperfect Time Series

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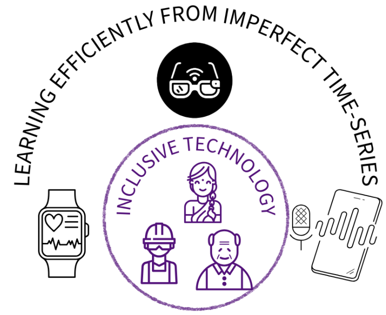
Wearables, smartphones, and continuous sensing applications are ubiquitous today. While they have achieved tremendous success through improved computation, smaller form factors, and advanced sensing capabilities, the analytics for these applications still lag behind those in the language and vision domains. This disparity arises because real-world time series data are *imperfect* — generally noisy, abruptly encountering missingness, and non-interpretable in raw format, among other challenges. Through my research, I aim to address these challenges in audio and sensing applications, particularly: 1) the lack of data and labels, 2) the dynamic nature of modalities—signal nonstationarity and missingness, and 3) factors for human-centered computation in sensing frameworks. I seek to tackle these issues with a focus on inclusivity and real-time deployability, going beyond mere performance enhancement. In one of my studies on fatigue monitoring, conducted with 45 subjects using real manufacturing setups and a wearable sensing framework, we found that the major challenge is the subjective nature of perceived fatigue scores among participants. This requires an ordinality-aware modeling technique and must be lightweight to provide real-time actionable feedback to operators on actual factory floors. To overcome data unreliability, including missingness of some modalities and labels for minority audio applications such as dysfluency detection, I have designed methods using self-supervised learning and resilient multimodal architectures. I also systematically design algorithms to address signal nonstationarity with minimal design overhead across a wide range of time-series sensing applications. Continuous health monitoring devices stand to benefit directly from such dedicated, data-driven frameworks. Ultimately, my vision is to propel sensing technologies from being merely *smart*—capable of data collection—to truly *intelligent*, where they analyze data seamlessly and deliver meaningful, actionable insights. These intelligent systems will blend effortlessly into daily life, enhancing human experiences while being equitable.



Research Vision : Intelligent and Inclusive Sensing Frameworks.